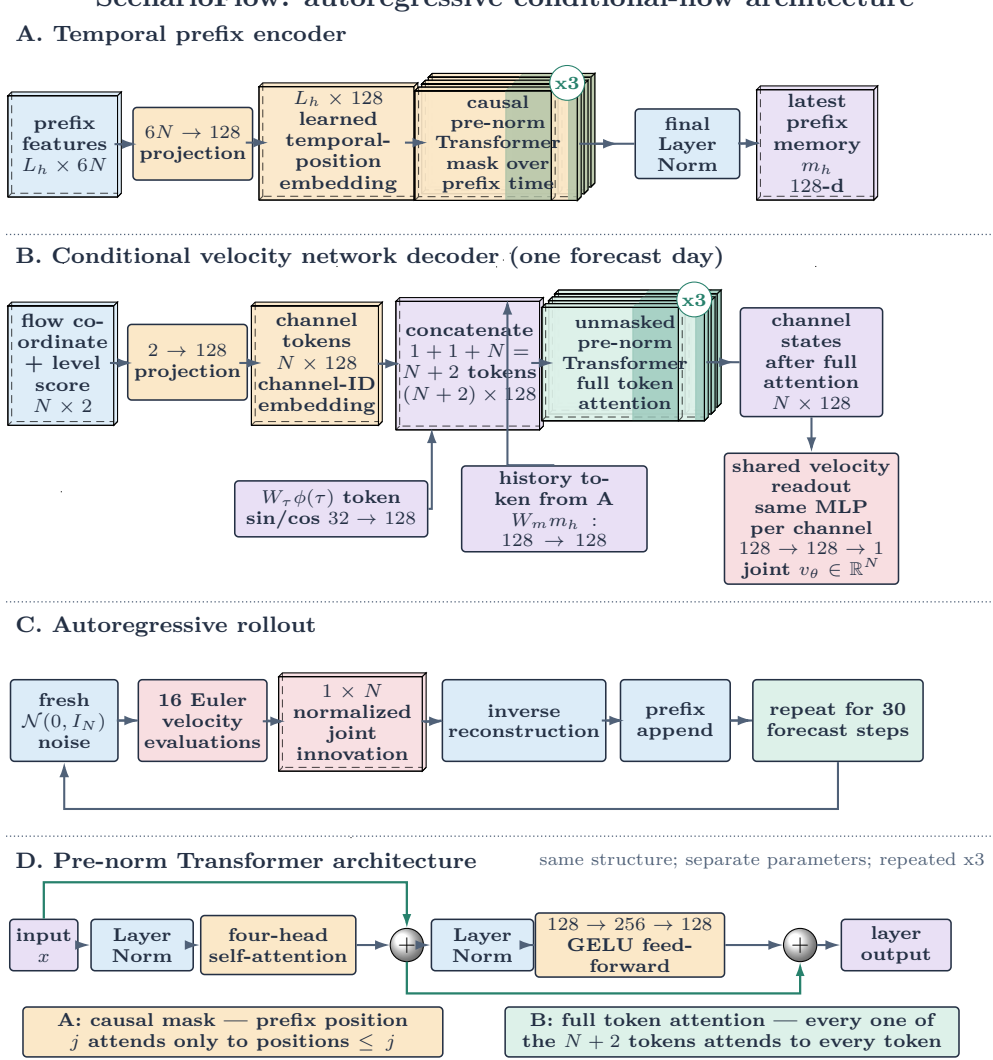

ScenarioFlow: autoregressive conditional-flow architecture
A. Temporal prefix encoder
prefix features $L_h \times 6N$
$6N \to 128$ projection
$L_h \times 128$ learned temporal-position embedding
causal pre-norm Transformer mask over prefix time
x3
final Layer Norm
latest prefix memory m_h 128-d
B. Conditional velocity network decoder (one forecast day)
flow co-ordinate + level score $N \times 2$
$2 \to 128$ projection
channel tokens $N \times 128$ channel-ID embedding
concatenate $1 + 1 + N = N + 2$ tokens $(N + 2) \times 128$
unmasked pre-norm Transformer full token attention
x3
channel states after full attention $N \times 128$
$W_\tau \phi(\tau)$ token sin/cos 32 $\to$ 128
history token from A $W_m m_h$: 128 $\to$ 128
shared velocity readout same MLP per channel $128 \to 128 \to 1$ joint $v_\theta \in \mathbb{R}^N$
C. Autoregressive rollout
fresh $\mathcal{N}(0, I_N)$ noise
16 Euler velocity evaluations
$1 \times N$ normalized joint innovation
inverse reconstruction
prefix append
repeat for 30 forecast steps
D. Pre-norm Transformer architecture
same structure; separate parameters; repeated x3
input x
Layer Norm
four-head self-attention
Layer Norm
$128 \to 256 \to 128$ GELU feed-forward
layer output
A: causal mask — prefix position j attends only to positions $\leq j$
B: full token attention — every one of the $N + 2$ tokens attends to every token